

WRITING LAB REPORTS

– A FIRST YEAR GUIDE

Three general objectives are to be accomplished in Physics labs. First, is to gain insights into a variety of Physics principles. Second, is to learn something about the process of scientific investigation and verification. Finally, we need to learn how to communicate what we know or have seen. Your lab report is the vehicle by which we can find out how well these objectives have been accomplished. To meet these objectives the student must know:

1. How to use various measuring instruments and to estimate their reading errors
2. How to reduce/eliminate errors
3. How to calculate errors
4. How to use various common instruments
5. Methods of analysis and how to draw and interpret linear and curved graphs
6. To discuss results and draw intelligent conclusions
7. Laboratory technique and laboratory safety

STRUCTURE OF A LAB REPORT

(The whole report should be at least 15 pages (excluding graphs and drawing) hand written with minimum spacing between lines and words.

Title

The title of a lab should be related to what was done in the lab. Its primary purpose is to make sure one does not lose track of what the report would be about.

Objectives

Why was the practical done? What was the student trying to achieve? The objectives should be listed and numbered. This will help in keeping the student in line and also important when drawing the conclusion. A report without objectives cannot be considered complete.

Theory /Literature review

This section sets the background that the reader will need to understand the rest of the lab report. Here you should discuss the Physics that the reader will need to know including equations if any. Explain the specific concepts, laws/formula, and theories that are relevant to the experiment. This means explaining

the terms in the formula and how they are present or experienced in the experiment. This section helps clarify the ideas that one needs to apply to the specific experiment at hand. **(Make this at least two(2) pages)**

Experimental procedure/Method

A detailed but concise discussion of the experimental setup, how you calibrated your measuring tools and how you took your data must be stated in this section. A list of apparatus and sometimes diagram of the apparatus and experimental setup/design is often given here.

Important to note is that the procedure should be in reported speech. You are reporting what you did and not giving instructions on what should be done. Never write in first person. The following are examples on how to write the procedure:

- a) “The ball was dropped from a known height and the time it took to travel to a designated distance was then noted”
- b) “We dropped the ball from a known height and noted the time it took to travel to a designated distance”

This is a very important part of any report and students should take it as such.

Data Presentation and Discussion

In this section (**the most important part of the lab/report**), you present your data, usually in graphical or/and **tabular form**, and present a **discussion** which interprets the results for the reader. Experimental values that are not consistent should not be included. The narrative of this section must lead the reader to the same conclusion as you reached rather than allow the reader to reach their own conclusions. When presenting data you must remember to include the correct number of significant figures.

Calculations fall in this category for experiments that need calculations. These should be well detailed, procedural and correct. Error calculation and discussion on how to correct them are also made here. The student must explain and discuss anything that is in the experiment that they wish the reader to know.

If done well, this section will lead one to the obvious conclusion of the experiment.

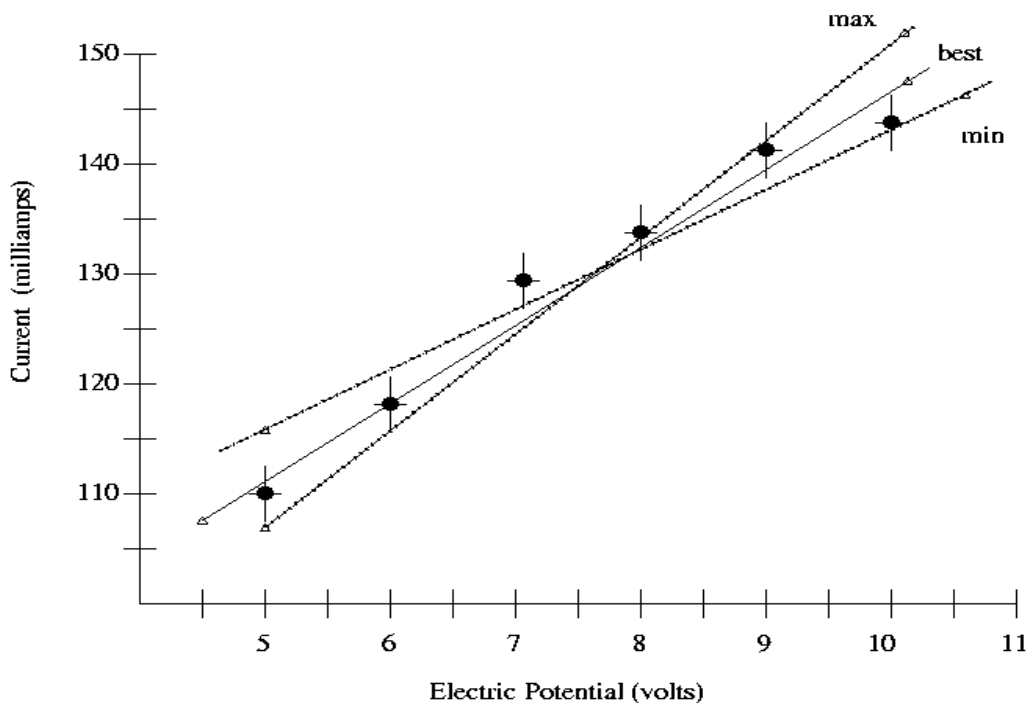
Error analysis:

This is usually the uncertainty of the results. The analysis is usually derived from the reading error of the raw data taken from the lab. There are many kinds of errors like random errors, systematic errors, absolute errors and so on.

- a) In most of the experiments, the initial readings are used to calculate out some quantity **x**. This will have the same units the same as the quantity itself.
- b) As the fractional error, dx/x .
- c) As percentage error = $(dx/x) \times 100\%$. Detailed error analysis should be got from error analysis books, internet and also from class lecture notes.

d) Uncertainty in the slope on a graph

If one has more than a few points on a graph, one should calculate the uncertainty in the slope as follows. In the picture below, the data points are shown by small, filled, black circles; each datum has error bars to indicate the uncertainty in each measurement. It appears that current is measured to ± 2.5 milliamps, and voltage to about ± 0.1 volts. The hollow triangles represent points used to calculate slopes. Notice how I picked points near the ends of the lines to calculate the slopes!



1. Draw the "best" line through all the points, taking into account the error bars. Measure the slope of this line.
2. Draw the "min" line -- the one with as small a slope as you think reasonable (taking into account error bars), while still doing a fair job of representing all the data. Measure the slope of this line.
3. Draw the "max" line -- the one with as large a slope as you think reasonable (taking into account error bars), while still doing a fair job of representing all the data. Measure the slope of this line.
4. Calculate the uncertainty in the slope as one-half of the difference between max and min slopes.

In the example above, find

$$\begin{aligned} \text{"best" slope} &= \frac{147 \text{ mA} - 107 \text{ mA}}{10 \text{ V} - 4.5 \text{ V}} = 7.27\Omega \\ &\quad 145 \text{ mA} - 115 \text{ mA} \end{aligned}$$

$$\text{"min" slope} = \frac{\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad}{10.5 \text{ V} - 5.0 \text{ V}} = 5.45\Omega$$

$$\text{"max" slope} = \frac{152 \text{ mA} - 106 \text{ mA}}{10 \text{ V} - 5.0 \text{ V}} = 9.20\Omega$$

$$\text{Uncertainty in slope is } 0.5 * (9.20 - 5.45) = 1.875\Omega$$

There are at most two significant digits in the slope, based on the uncertainty. So, I would say the graph shows

$$\text{Slope} = 7.3 \pm 1.9\Omega$$

Source of errors:

There is usually no ideal results. The reality is that some physical restraints that affect the actual performance of the experiment. In this section please indicate these physical restraints.

Precautions:

These are measures that you should take to protect the safety of oneself and the equipments.

Recommendations:

The part should entail how best the experiment should be done in order to achieve the desired aim

Conclusions

Draw a logical conclusion based on your results. The conclusions should answer the questions posed in the objectives. If one was to read your conclusions they must see that you have responded to the set objectives and answered them.

References

State your sources of information that assisted you in compiling the lab. One must have referred somewhere.

In writing the references we ask that you use the **APA STYLE OF REFERENCING**. The order in which the reference is written is very important. Read and understand on how to reference using the APA format.

GENERAL COMMENTS

1. Each lab report must be written in English, following the usual rules of grammar and spelling. Be sure that you use the proper punctuation, so that the narrative will have some logical flow. Lab write-ups should be done in persuasive manner. You want to

- tell the reader what to see in your data, lead the reader to the conclusions. Do not just present a bunch of data and expect the reader to understand what it is all about. The reader should finish reading the lab having the same interpretation of data as you do.
2. The lab write-up is not a journal of what you did and your thoughts while doing this experiment. You do not discuss your frustrations or wrong paths you followed while you were moving to an understanding of the experimental results.
 3. Each figure or table needs to be numbered and have a caption with it. A figure or table that is not discussed in the narrative section of the lab report is not needed and should be eliminated. Figures and tables should be referred to by their number in the text e.g. Table 2 shows the data...etc.
 4. Labs should be written in third person and in the past tense; “the index of refraction of glass was found by measuring the angle of deflection of...” OR “We found the index of refraction of glass by measuring the angle of deflection of...” (Even if only one person did the lab!)
 5. Formulae should be presented on separate lines and numbered sequentially through the report. You should refer to a formula its number e.g. ...substituting Equation 3 into Equation 4 yields...
 6. Do not forget to write your name, class, date, registration number and any other pertinent identifying information.
 7. Grading: the lab reports will contribute 10 marks towards your final grade in the respective units, and missing 2/3 of the lab work automatically disqualifies a student from proceeding with the academic program in the University.
 8. Basically your report should be divided into the following topics: Basically your report should be divided into the following topics: Personal details page, Title, Objectives, Literature Review, Apparatus/reagents/equipment, Method/procedure, table of Results, data analysis and discussion, , error analysis, source of errors, precautions, Conclusion, and Reference.
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